UNIVERSITY OF ENGINEERING & TECHNOLOGY MARDAN

COMPUTER SCIENCE

CS-305L: Parallel & Distributed Computing Lab

SUBMITTED BY

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REG NO : 23MDBCS 435

SECTION : CS-A

SEMESTER : 6th

Lab 2

1. Explain the role of mpirun/ mpiexec and why we use it to run MPI programs.

`mpirun` or `mpiexec` is a command used to execute MPI programs in parallel. It starts multiple instances of the same program at the same time, where each instance runs as a separate process. Each process is assigned a unique rank (process ID).

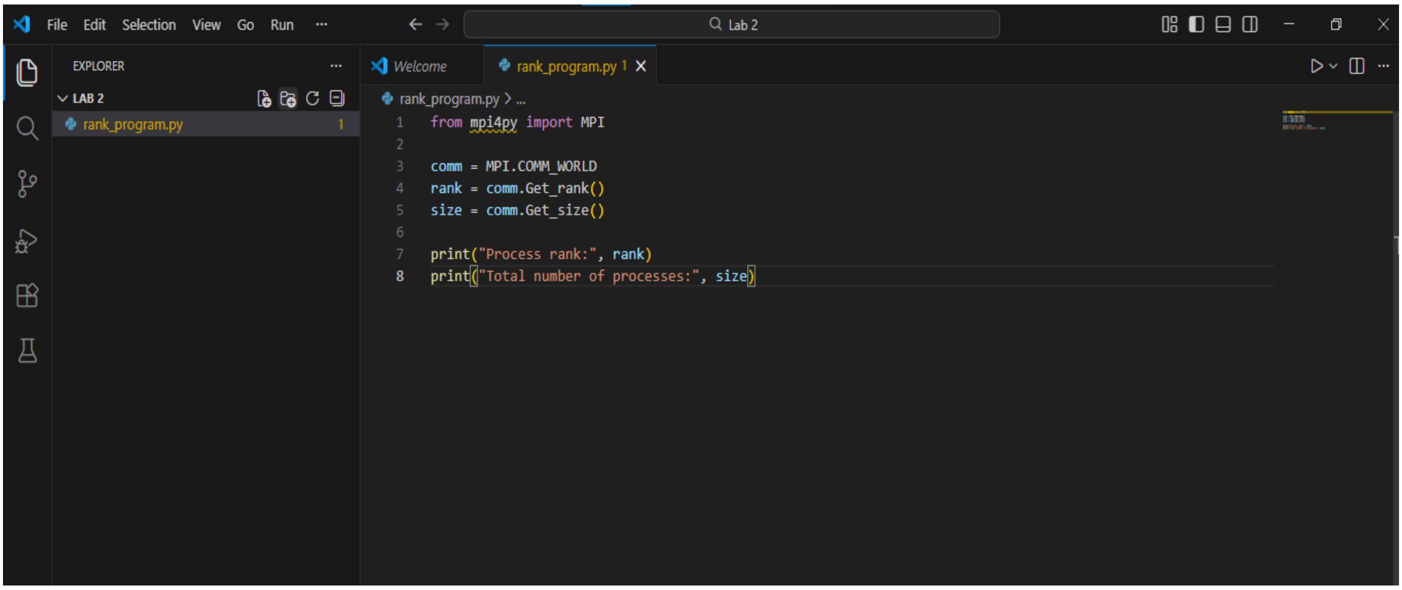
We use `mpiexec` because a normal Python execution runs only one process, while MPI programs require multiple processes to perform parallel computation.

Example:

```
mpiexec -n 4 python program.py
```

This command runs the program using four parallel processes.

2. Write a Python script that retrieves and prints the rank (process ID) and total number of processes.



Type in Anaconda Promt

```
(mpi_env) C:\Users\Rifaq Ajmal>conda activate mpi_env

(mpi_env) C:\Users\Rifaq Ajmal>cd Desktop\Lab_1
The system cannot find the path specified.

(mpi_env) C:\Users\Rifaq Ajmal>mpiexec -n 4 python rank_program.py
```

Output

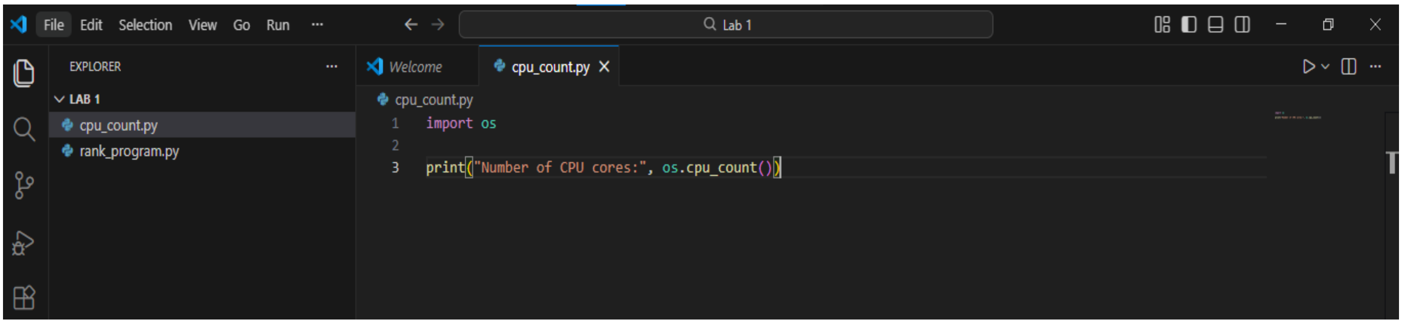
```
(mpi_env) C:\Users\Rifaq Ajmal\Desktop\Lab_1>mpiexec -n 4 python rank_program.py
Process rank: 1
Total number of processes: 4
Process rank: 3
Total number of processes: 4
Process rank: 0
Total number of processes: 4
Process rank: 2
Total number of processes: 4

(mpi_env) C:\Users\Rifaq Ajmal\Desktop\Lab_1>
```

3. Explain what MPI.COMM_WORLD does in an MPI program.

MPI.COMM_WORLD is the default communicator in MPI that includes all processes running in an MPI program. It allows processes to communicate with each other and helps each process know its rank and the total number of processes. It connects all processes so they can work together in parallel.

4. Write a program Count the number of CPU cores installed on your system using Python.



Output

```
(mpi_env) C:\Users\Rifaq Ajmal\Desktop\Lab_1>python cpu_count.py
Number of CPU cores: 4

(mpi_env) C:\Users\Rifaq Ajmal\Desktop\Lab_1>
```